HMGP 7622

Rethinking data analysis

A researcher's guide to avoiding missteps and misuse

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Day 05

Pseudoreplication and Confounding

- Replication & sources of variation
- Pseudoreplication: getting the experimental unit wrong
- Effective sample size
- Confounding variables: the alternative explanation
- Batch effects and technical artifacts

Replication and sources of variation

Science relies on replicate measurements

- Var(measurement) = External factor variability

  + Natural biological variability

  + Measurement error

- Additional replicates → more accurate & reliable summary statistics.
- Replicates can be used to:
  - Assess & isolate sources of variation in measurements
  - Limit the effect of spurious variation on hypothesis testing & parameter estimation.

Blainey et al., (2014) *Nat. Meth.*

Replication and sources of variation

Biological & technical replicates

Biological replicates:

- Parallel measurements of biologically distinct samples
- Capture random biological variation (could be subject of study or a noise source).

Technical replicates:

- Repeated measurements of the same sample
- Represent independent measures of the random noise associated with protocols or equipment.

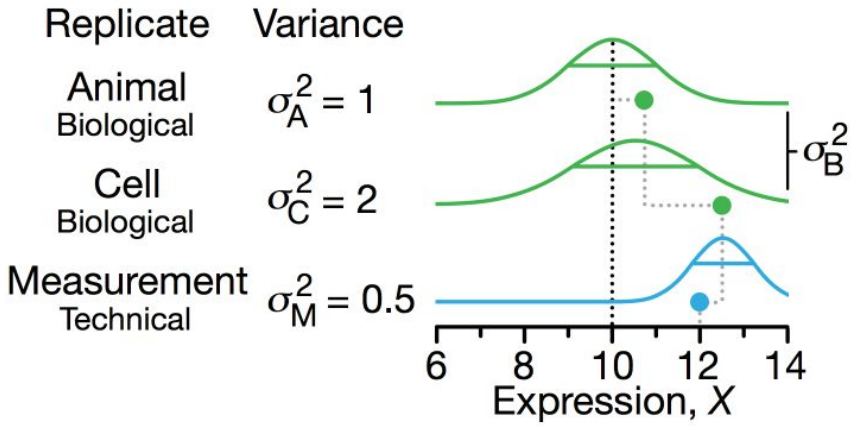
Which sources of variation are being studied & which are considered noise?
B: biological, T: technical

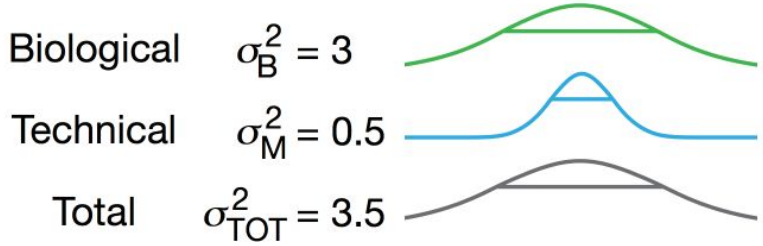
	Replicate type	Replicate category[a]
Animal study subjects	Colonies	B
	Strains	B
	Cohoused groups	B
	Gender	B
	Individuals	B
Sample preparation	Organs from sacrificed animals	B
	Methods for dissociating cells from tissue	T
	Dissociation runs from given tissue sample	T
	Individual cells	B
	RNA-seq library construction	T
Sequencing	Runs from the library of a given cell	T
	Reads from different transcript molecules	V[b]
	Reads with unique molecular identifier (UMI) from a given transcript molecule	T

Blainey et al., (2014) *Nat. Meth.*

Replication and sources of variation

Biological & technical replicates





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Effective sample size

Dependency between observations violates the tenet that data should be independent and identically distributed (IID). Correlated observations represent redundant information.

Even small levels of Intraclass Correlation Coefficient (ICC) (correlation within a group) can have a massive impact.

If ρ is the correlation between samples,

$$n_{\text{eff}} = \frac{n}{1 + (n-1)\rho}$$

The denominator: Variance Inflation Factor (VIF)

$n_{\text{eff}} \neq n$: Pseudoreplication

Sample size (n)

$$\text{Var}(\bar{X}) = \sigma^2 / n$$

Effective sample size (n_{eff})

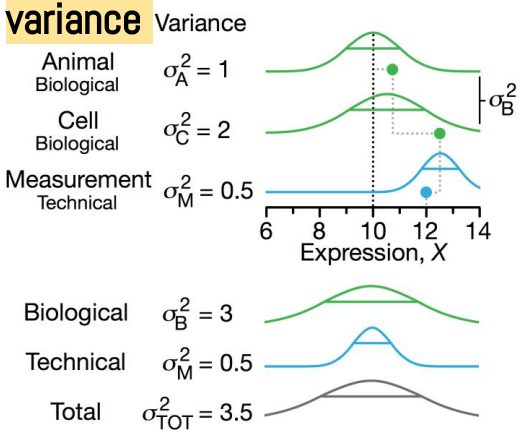
$$\text{Var}(\bar{X}) = \sigma^2 / n_{\text{eff}}$$

E.g., in one trial, a deceivingly small ICC of 0.03 required an 8.5-fold increase in the number of participants to maintain the same statistical power as an independent sample.

Blainey et al., (2014) *Nat. Meth.*

Pseudoreplication: Getting the experimental unit wrong

Impact of replication number on variance



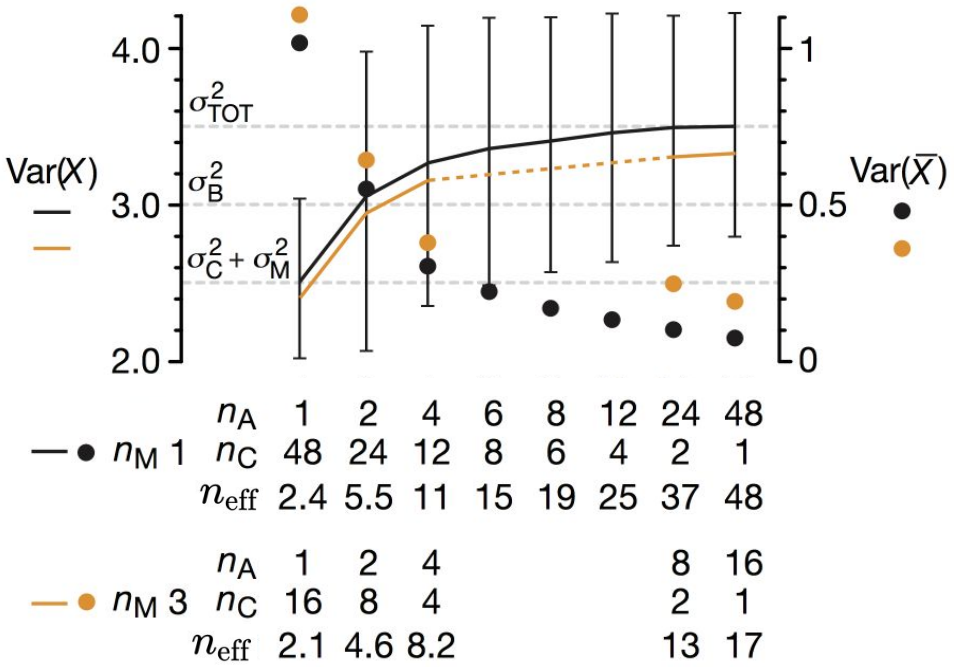
Simulation

$n = n_A n_C n_M = 48$

$n_A = 1{:}48,\ n_C = 1{:}48,\ n_M = 1, 3$

$n_{eff} = 2{:}48 = \mathrm{Var}(X)/\mathrm{Var}(\bar{X})$

Variance of expression & expression mean



$\mathrm{Var}(\bar{X}) = \sigma_A{}^2/n_A + \sigma_C{}^2/n_A n_C + \sigma_M{}^2/n_A n_C n_M$

Blainey et al., (2014) *Nat. Meth.*

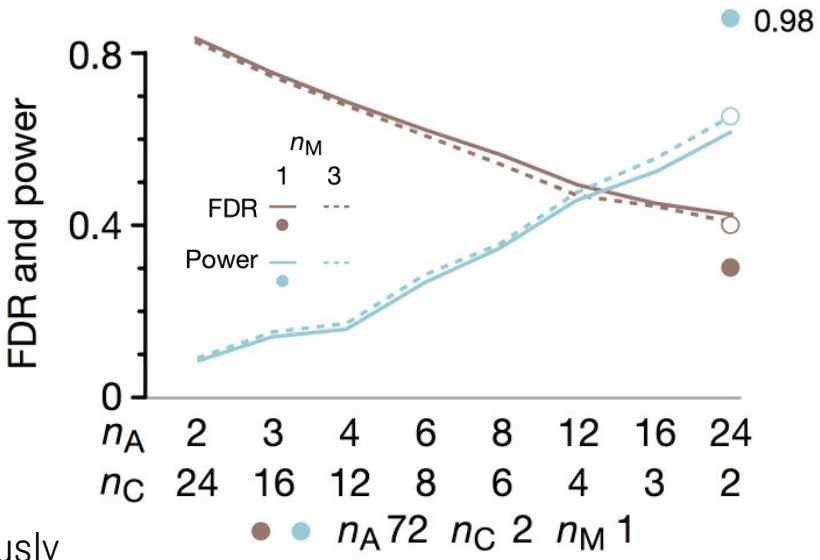
Pseudoreplication: Getting the experimental unit wrong

Impact of r eplication number on inference

Simulation of 10% effect in mean

- More animals the better.
- (n_A, n_C, n_M) from (24, 2, 3) to (72, 2, 1): 50% inc. in power (0.66→0.98).

Inference of difference in means



But, consider cost difference b/w biological & technical replicates.

Mixing within- & between-subject analysis units spuriously increases the degrees of freedom
→ much easier to cross "significance" threshold
→ results that appear robust but are actually wildly optimistic or purely the result of noise

Blainey et al., (2014) *Nat. Meth.*

Pseudoreplication: Getting the experimental unit wrong

Patterns

- **Nested data:** Students within one classroom, patients within one clinic, or multiple cells from a single culture
- **Repeated measurements:** Treating the pre-score and post-score of a single subject as independent points in a regression
- **Litter effects:** In animal studies, if the mother is the one treated with a drug, the mother is the experimental unit, not the individual fetuses in her litter

Lookout for reported n values that reflect the number of observations rather than the number of independent subjects. Papers often claim high n (e.g., n=100 cells) when the effective n (e.g., n=12 animals) is far lower, rendering the resulting p-values meaningless.

Pseudoreplication: Getting the experimental unit wrong

Some takeaways

Typically, biological variability >> technical variability.

- Commit resources to sampling biologically relevant variables (unless measures of technical variability are themselves of interest).

Planning for replication:

1. Identify the question the experiment aims to answer.
2. Determine proportion of variability induced by each step.
3. Distribute the capacity for replication of the experiment across steps.
4. Be aware of the potential for pseudoreplication and aim to design statistically independent replicates.

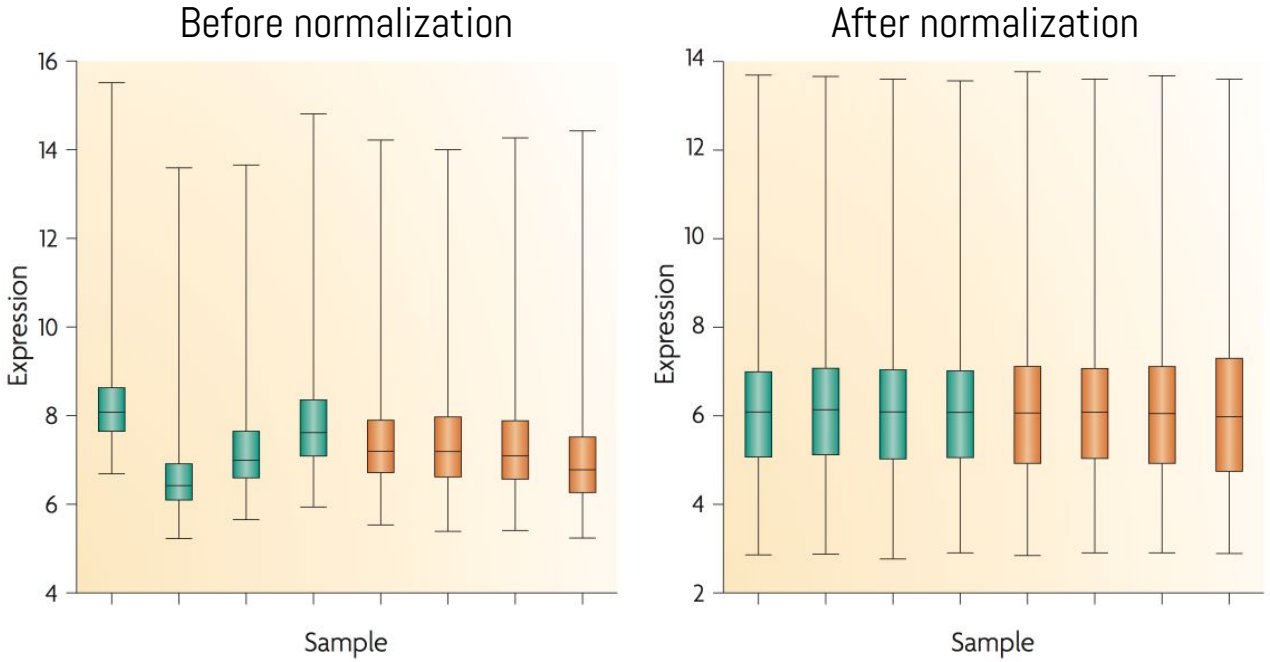
Hierarchical/mixed models: These are usually necessary for clustered data, allowing researchers to define variability within subjects as a fixed effect and between-subject variability as a random effect.

As capacity for higher-throughput assays increases: more is not always better.

Blainey et al., (2014) *Nat. Meth.*

Confounding variables

Extraneous variables (e.g. processing data) can be *confounded* with the outcome of interest (e.g. disease state) when it correlates both with the outcome and with an independent variable of interest (e.g. gene expression).



Leek et al., (2010) *Nat. Rev. Genet.*

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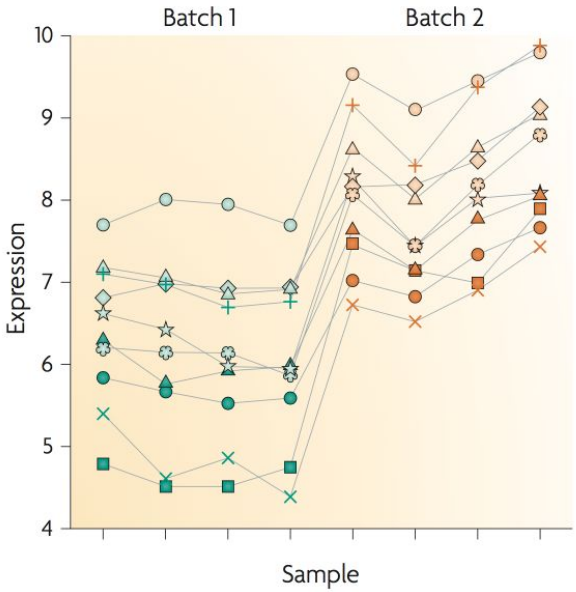
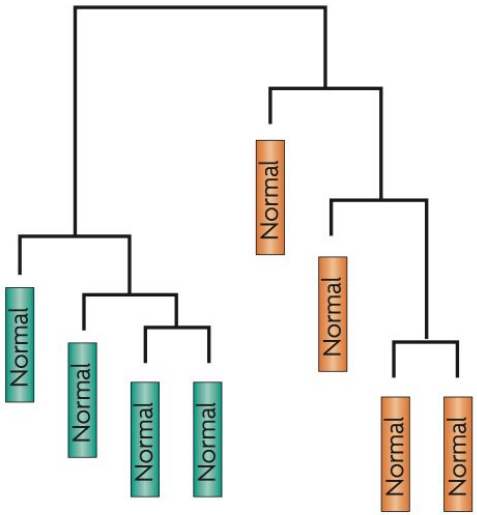
After normalization



Clustering of samples



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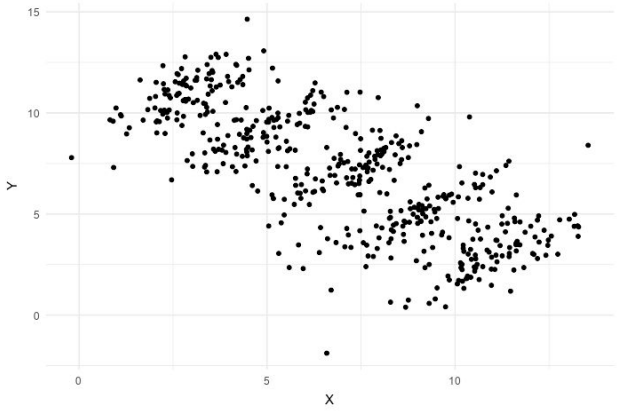
Confounding variables

Common scenarios

Simpson's paradox: A trend can appear in different subsets of data but reverse or disappear when the groups are combined due to a "lurking variable" like age or baseline health.

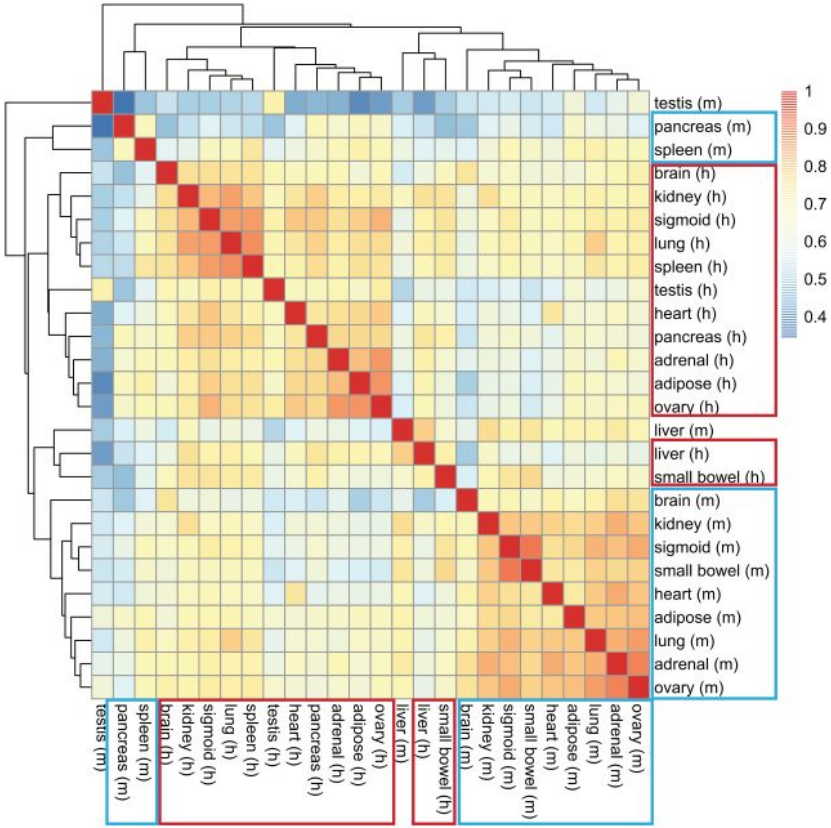
Selection bias: For e.g., if the control and treatment groups differ significantly in baseline values (e.g., age or bone density) before an intervention, the results are "doomed before they start".



Treatment / Stone size	Open	Closed
Small stones	*Group 1* **93% (81/87)**	*Group 2* 87% (234/270)
Large stones	*Group 3* **73% (192/263)**	*Group 4* 69% (55/80)
Both	78% (273/350)	**83% (289/350)**

Leek et al., (2010) *Nat. Rev. Genet.*

Confounding variables



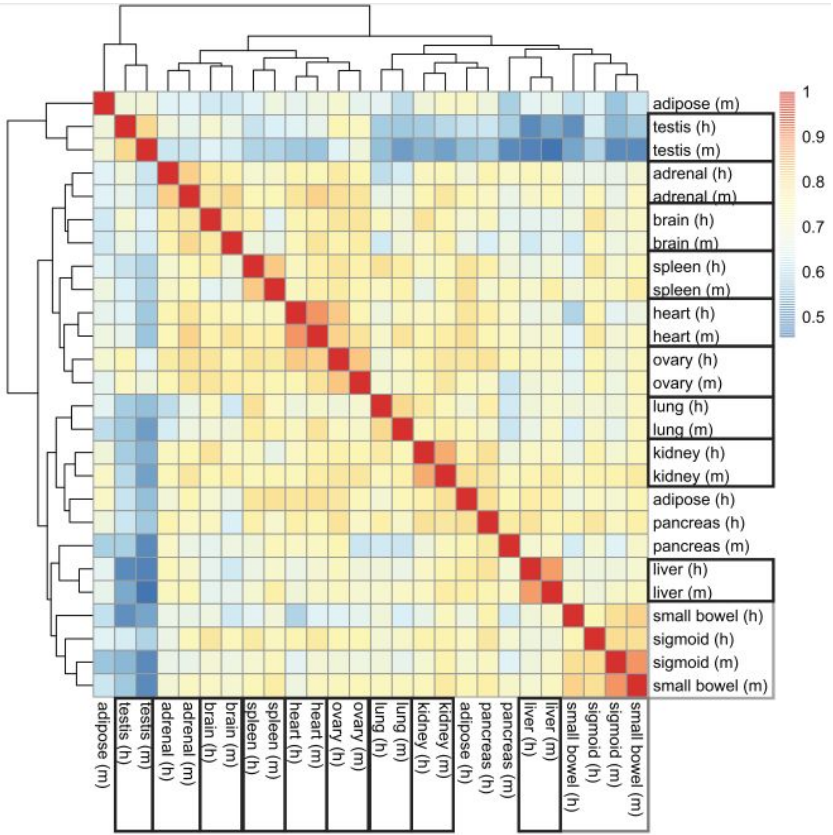
Mouse ENCODE comparative gene expression data

Lin et al. (2011) Comparison of the transcriptional landscapes between human and mouse tissues. PNAS 111:17224.

D87PMJN1 (run 253, flow cell D2GUAACXX, lane 7)	D87PMJN1 (run 253, flow cell D2GUAACXX , lane 8)	D4LHBFN1 (run 276, flow cell C2HKJACXX , lane 4)	MONK (run 312, flow cell C2GR3ACXX , lane 6)	HWI-ST373 (run 375, flow cell C3172ACXX , lane 7)
heart	adipose	adipose	heart	brain
kidney	adrenal	adrenal	kidney	pancreas
liver	sigmoid colon	sigmoid colon	liver	brain
small bowel	lung	lung	small bowel	spleen
spleen	ovary	ovary	testis	● Human
testis		pancreas		● Mouse

Gilad & Mizrahi-Man (2015) *F1000Res*

Confounding variables



Re-analysis after correcting for batch-effects.

Gilad et al. (2015) A reanalysis of mouse ENCODE comparative gene expression data. F1000Res 4:121.

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Confounding variables

Exploratory analyses

Hierarchically cluster the samples and label them with biological variables and batch surrogates (such as laboratory and processing time)

↓

Plot individual features versus biological variables and batch surrogates

↓

Calculate principal components of the high-throughput data and identify components that correlate with batch surrogates

Leek et al., (2010) *Nat. Rev. Genet.*

Confounding variables

Downstream analyses

Do you believe that measured batch surrogates (processing time, laboratory, etc.) represent the only potential artefacts in the data?

Yes ↓ Use measured technical variables as surrogates for batch and other technical artefacts

No ↓ Estimate artefacts from the high-throughput data directly using surrogate variable analysis (SVA)

↓

Perform downstream analyses, such as regressions, t-tests or clustering, and adjust for surrogate or estimated batch effects. The estimated/surrogate variables should be treated as standard covariates, such as sex or age, in subsequent analyses or adjusted for use with tools such as ComBat

Diagnostic analyses

Use of SVA and ComBat does not guarantee that batch effects have been addressed. After fitting models, including processing time and date or surrogate variables estimated with SVA, re-cluster the data to ensure that the clusters are not still driven by batch effects

Leek et al., (2010) *Nat. Rev. Genet.*